

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the Software Development Engineer with Amazon Web Services. I bring over three years of professional experience in full-stack web applications, RAG and AI agents, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade backend systems that directly reflect the complex software solutions and distributed systems this position demands. As a Software Engineer at Sperse, I designed and developed a large-scale, multi-tiered MCP AI interaction layer that translates natural-language intents into safe, validated tool calls across 1000+ platform endpoints — requiring rigorous API contract design, input validation, and safe execution across a distributed microservices surface. To keep these systems production-ready, I built a robust observability layer using Langfuse and LangSmith for tracing, structured logs, and evaluation hooks, reducing end-to-end latency by 60% and lowering API costs by 50% by identifying and resolving performance bottlenecks through optimized prompt routing and agent concurrency. Prior to Sperse, as a Software Engineer at Qualitest, I contributed to the backbone infrastructure that thousands of users rely on daily. I designed and developed a distributed systems Storage Service using Python, PostgreSQL, and Redis that seamlessly handles large-scale file operations in S3 and MinIO via expiring signed URLs — giving me direct, hands-on experience with cloud-native, multi-environment backend design. I also collaborated with cross-functional teams to implement an Identity & Access Management platform using React, Spring Boot, and AWS, supporting thousands of users across 30+ applications, and identified opportunities for process enhancement by automating long-running data import jobs using AWS Step Functions, preventing AWS Lambda timeouts and debugging production issues — both of which required the kind of systems-level thinking and operational stability focus this role demands. Together, these roles have prepared me to design, develop, and maintain efficient, reusable, and reliable code and drive impact within an agile engineering team.

My academic research experience has given me a rigorous foundation in model development, GPU-accelerated computing, and technical communication — skills that translate directly into the passion for invention and commitment to operational excellence this team values. As an AI Researcher at the Pace Artificial Intelligence Lab, I implemented training/inference lifecycles for StyleGAN and Stable Diffusion utilizing CUDA and PyTorch on an HPC cluster, applying optimization techniques for Machine Learning workloads. I was responsible for managing the full model development lifecycle — from data preparation and training runs to utilizing vLLM for model inference with Hugging Face deployment — leveraging knowledge of LLM fundamentals and transformer architecture to ensure secure and efficient model serving. Beyond building the systems myself, I mentored junior engineers and +50 students on GPU programming with CUDA and Mojo, and on building AI Agents. Designing and delivering those sessions required me to break down complex technical concepts clearly and accessibly — directly mirroring the mentorship of junior engineers and contribution to a strong engineering culture central to this role. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, both of which I would bring to Amazon Web Services and its mission.

I have always been passionate about customer obsession and long-term thinking and envision myself contributing my talent to the furthering of building technology that empowers builders, organizations, and communities to solve the world's hardest problems. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan